Introduction to Linear Algebra

Vectors, Matrices, and Beyond

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What We'll Cover

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- Applications in data science

Vector Spaces

Definition of Vector Space

A **vector space** V over field F has two operations:

Addition

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Together, these satisfy the 8 vector space axioms.

Examples of Vector Spaces

Euclidean Space $\mathbb{R}^n$

The most familiar vector space:

$$\mathbb{R}^3 = \{(x, y, z) : x, y, z \in \mathbb{R}\}$$

With standard addition and scalar multiplication.

Examples of Vector Spaces

Polynomial Space $\mathbb{P}_n$

Polynomials of degree at most n:

$$\mathbb{P}_2 = \{a_0 + a_1 x + a_2 x^2 : a_i \in \mathbb{R}\}$$

This is a 3-dimensional vector space!

Examples of Vector Spaces

Function Spaces

Continuous functions $C[a, b]$ form an **infinite-dimensional** space:

$$C[0, 1] = \{f : [0, 1] \to \mathbb{R} : f \text{ is continuous}\}$$

Linear Independence

A key concept for bases

Vectors $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_n$ are **linearly independent** if:

$$c_1 \boldsymbol{v}_1 + c_2 \boldsymbol{v}_2 + \ldots + c_n \boldsymbol{v}_n = \mathbf{0}$$

implies $c_1 = c_2 = \ldots = c_n = 0$.

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Example: In $\mathbb{R}^2$, the vectors $\binom{1}{0}$ and $\binom{0}{1}$ are linearly independent.

But $\binom{1}{2}$ and $\binom{2}{4}$ are **not** (one is a scalar multiple of the other).

Linear Transformations

The Rank-Nullity Theorem

Theorem *Rank-Nullity.*

For linear $T : V \to W$ with V finite-dimensional:

$$\dim(\ker T) + \dim(\operatorname{im} T) = \dim(V)$$

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Intuition: The dimensions you “lose” (kernel) plus what you “keep” (image) equals what you started with.

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Surjective (Onto)

$\operatorname{im}(T) = W$

- Every output is hit
- "Covers" the codomain
- $\dim(\ker T) = \dim(V) - \dim(W)$

Eigenvalues & Eigenvectors

Eigenvalue Equation

Definition *Eigenvalue/Eigenvector.*

For matrix A, scalar λ is an **eigenvalue** with **eigenvector** $\boldsymbol{v} \neq \mathbf{0}$ if:

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To find eigenvalues, solve:

$$\det(A - \lambda I) = 0$$

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Eigenvalues: $\lambda_1 = 3$ and $\lambda_2 = 2$

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- **PCA:** Eigenvectors of covariance matrix
- **Quantum Mechanics:** Observable eigenvalues
- **Differential Equations:** Stability analysis

Summary

Vector Spaces	Sets with addition & scalar multiplication
Linear Maps	Structure-preserving transformations
Eigenvalues	Scaling factors for special directions
Rank-Nullity	$\dim(\ker) + \dim(\mathrm{im}) = \dim(V)$

Thank You!

Questions?

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Slides: Made with axiomst